# INDIAN OIL CORPORATION LIMITED
**(Pipelines Division, NOIDA)**

| INTER OFFICE MEMO | | | |
|---|---|---|---|
| **From** | **DGM(MNT), PLHO, Noida** | **To** | **DGM (T) NRPL Panipat** |
| **Ref** | **PLHO/MNT/10.0** | **Ref** | |
| **Date** | **2.6.2015** | **Date** | |

**Sub: Investigation report on Breakdown of 500 KVA DG Set at SMPL Rewari**

Subject investigation report is attached herewith; report consists of Probable Reasons of Breakdown, Actions for Restoration of the DG set and Suggestions for Improvement and Operation Practices. Suggestions of the committee are as under

- To carry out all maintenances as per schedule.
- Feedback of DG running status to be made available in the SCADA.
- Keeping the protective devices in healthy condition by simulation for tripping of DG set for both Lube oil pressure low and jacket water temperature high tripping from AMF panel and the same to be recorded in maintenance report format.
- During continuous running of DG, parameters (Lube oil pressure, Jacket water temperature, RPM, load current etc.) to be recorded on hourly basis.
- Technical literature of DG set and AMF panel to be arranged and made available in the station.

Apart from above, comments of Director (PL) may also be looked into. Action taken report may please forwarded to this office for further management apprisal.

This is for your information and further needful action, please.

2/6/15

(T.K.Bera)

**Encl**: As mentioned above

Cc: ED (O) PLHO Noida
GM (M&I, H, S & E) PLHO Noida
GM (T) NRPL Panipat

# INVESTIGATION REPORT ON BREAKDOWN OF 500 KVA DG SET AT SMPL, REWARI



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# INCIDENT INVESTIGATION REPORT ON FAILURE OF THE 500 KVA DG SET AT SMPL, REWARI

On 13.02.2015, mainline engine (MP3) operation was continued at SMPL Rewari at night to meet the increased throughput demand. At 21:30 hrs, the grid power supply failed and to continue engine operation, the 500 KVA Volvo Penta DG set was started. At 04:05 hrs. on 14.02.2015, the DG set tripped with splashing of lube oil and smoke coming out from engine. After preliminary inspection, the liner and engine block were reported damaged. The engine was not taking manual cranking and the crank shaft was reported to be jammed.

## Formation of Committee:

A committee, consisting of following officers, was constituted vide approved Note No. PL/HO/MNT/01 dated 14.02.2014 (**Flag A**) to investigate into the incident of failure of the DG set at SMPL, Rewari on 14.2.2015:

1. Mr. R.P. Pant, DGM (Maintenance), PLHO
2. Mr. Tribhuwan Kumar , SOM, NRPL, Panipat &
3. Mr. Rajesh Kumar Mishra, DPJM(T&I), PLHO

The scope of work assigned to the Committee is as under:

a) Identification of root cause of failure
b) Suggestion for improvement of maintenance and operation practices
c) Action Plan for early restoration of the DG set

## Visit of the Committee to Site:

The committee visited SMPL Rewari station on 16.02.2015 and investigated the incidence.

## Sources of Evidence:

- Statement of Shift in-charge Shri Mukesh, AO&ME, NRPL Rewari
- Log Book, Maintenance records, Historical record of DG set
- Information provided by maintenance personnel
- Inspection of DG set

## History of DG Set:

- The Volvo Penta make engine was procured vide PO No. RB/98139 dated 01.02.1999 and commissioned in May 1999 at SMPL Rewari after replacing old GRSE engine after coupling with existing AVK make Alternator(500 KVA).
- On 13.02.2015, due to power supply failure, the DG set was started at 21:30 Hrs. for starting MP3
- At 04:05 Hrs. on 14.2.2015, the DG set tripped with abrupt sound

- On inspection, lube oil was found splashed around DG set and smoke was observed coming out from the engine.
- On manual cranking, the engine was found jammed.
- The engine has run for 1671 cumulative running hours only.

**Specification of DG Set:**

Engine:

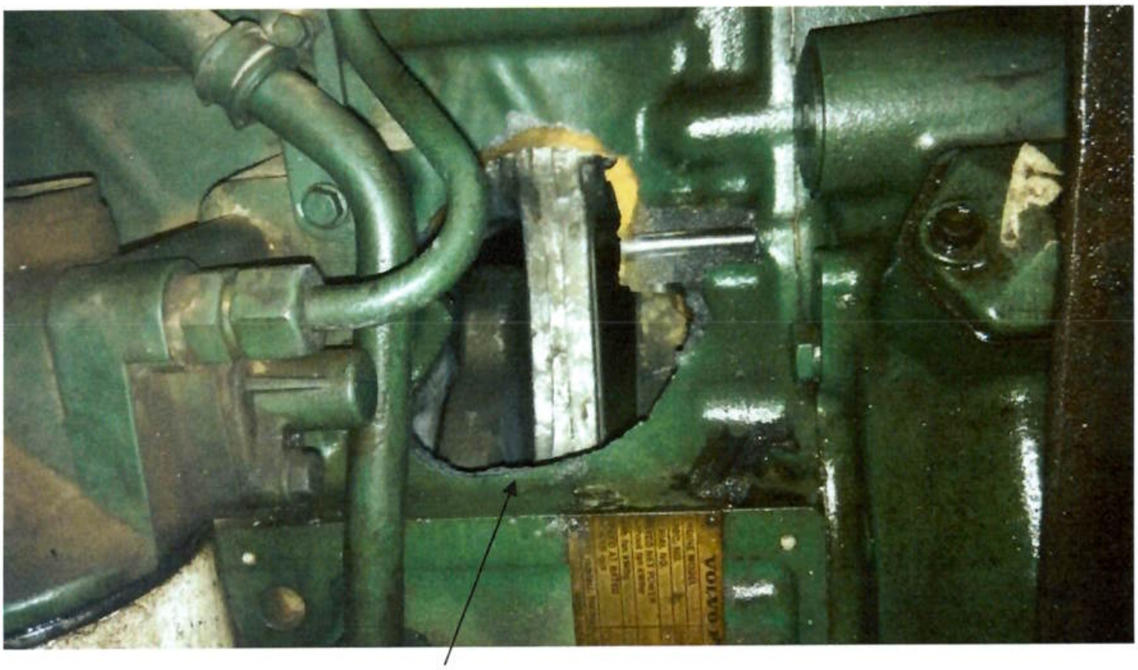
| | |
|---|---|
| Make | : Volvo Penta, Sweden |
| Type | : TAD 1631G |
| Specifications No. | : 868767 |
| No. of Cylinders | : 6 (Inline) |
| Sl. No. | : 2160028657 |
| Rated BHP | : 442 kw (Without fan) |
| Speed | : 1500 RPM |
| Bore | : 144 mm |
| Stroke | : 165 mm |
| Compression Ratio | : 15.0: 1 |
| Firing Order | : 1-5-3-6-2-4 |
| Lube Oil | : SP 15 W 40 |

Alternator:

| | |
|---|---|
| Make | : AVK |
| Rating | : 500 KVA |

**Observations after Dismantling of Cylinder Heads:**

- It was observed that a hole of 12 cm x 10 cm (Approx.) has been created in the engine block under Cylinder No. 6 (First from Alternator side) and have broken into several pieces.
- The top portion of Piston of Cylinder No. 6 found in pieces after melting.
- The bottom portion of the Piston was found stuck up with the Liner.



**Hole in the Cylinder Block**

**Engine Block after Removal of Cylinder Heads**

- Part of the Piston material was found stuck up on the bottom portion of Cylinder Head and Exhaust Valves.
- The top portion of Liner of Cylinder No. 6 was also found damaged. Pieces of Piston Ring were found with coating of white material of Piston.
- One Push Rod of Cylinder No. 6 found bent.
- Vertical scratch marks observed on the liners.
- Lube oil was found mixed with jacket water inside the sump. Lube oil is black in colour and with high viscosity even after running for only 174 hrs after last change.
- Lube oil starvation observed particularly on Liner No. 6 and 1.

**Pieces of Top Portion of Piston No. 6**

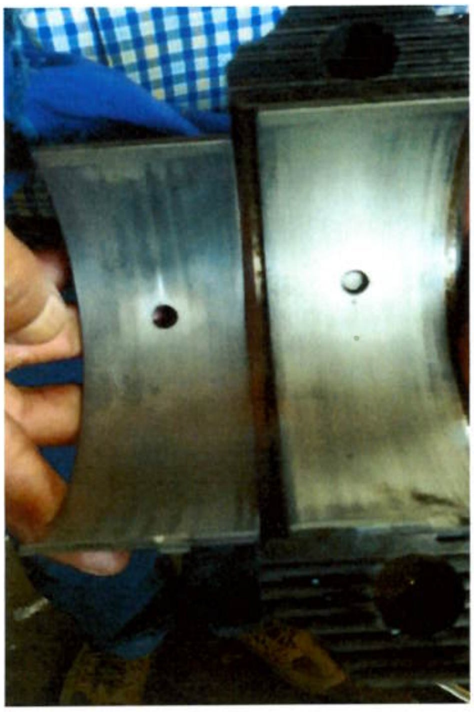
**Cylinder Head No. 6 with Valves**

**<u>Simulation of Alarms</u>:**

- Feedback of DG running status was not available in the SCADA.
- Operation of DG set was available in manual mode only from local AMF panel and auto mode was not available.
- Provision for tripping of DG in Lube oil pressure low and Jacket water temperature high alarm is provided in the panel.
- Lube oil pressure sensor and Jacket water temperature sensor were installed in the DG for monitoring and tripping purpose.
- Annual T&I maintenance for station DG operation & tripping check was carried out on 06.03.2014 as per scheduled maintenance plan and recorded in the IMS format no. QAF/T&I/I/19(**Flag B**).

**Vertical Scratch Marks Observed on Liners.**

**Exhaust Valves of Cylinder No. 6**



**Big End Bearing of Connecting Rod No. 6**

- As per the maintenance report, simulation for tripping on Jacket water temperature high was carried out at 93 Deg C.
- However, tripping check on Lube Oil pressure Low was not carried out.
- During visit of Committee the AMF panel was isolated, batteries were disconnected and jacket water temperature sensor was kept dismantled.
- NO/ NC contacts of Jacket water temperature switch was checked by calibration and change of state of contact was found OK.
- Lube Oil Pressure switch could not be checked because pressure calibrator was not available in the station.
- However, the committee checked the simulation for Lube oil pressure low and Jacket water temperature high tripping after re-energizing the AMF panel but the same was found not working.

**Maintenance History:**

Following is the maintenance history of DG set as recorded in the History Register:

- 30.11.2013: Half -Yearly maintenance done (No Maintenance Report available)
- 06.03.2014: Annual T&I maintenance done (IMS format No. QAF/T&I/I/19 available- **Flag B**).
- 18.09.2014: Lube oil (SP 15 W 40) and Diesel filters changed during monthly maintenance at 1497 running hours (No Maintenance Report available)
- 30.10.2014: Half -Yearly maintenance done (IMS format No. MMM:14 **-Flag C** )
- Lube Oil topped up on 19.11.2014, 22.12.2014 and 13.01.2015 (10 ltrs. each)
- Endurance test taken on 02.01.2015 for 6 hours. However, engine parameters not recorded.

As per the approved Annual Maintenance Schedule for 2014-15, following maintenances of the DG set engine were to be carried out:

| | |
|---|---|
| 250 Hrs./Monthly | : Every month |
| 1500 Hrs./Half Yearly/Yearly | : May 2014 (Yearly) |
| | : November 2014 (Half Yearly) |
| 3000 Hrs./ 2-Yearly | : May 2014 |
| 6000 Hrs./ 4-Yearly | : May 2014 |

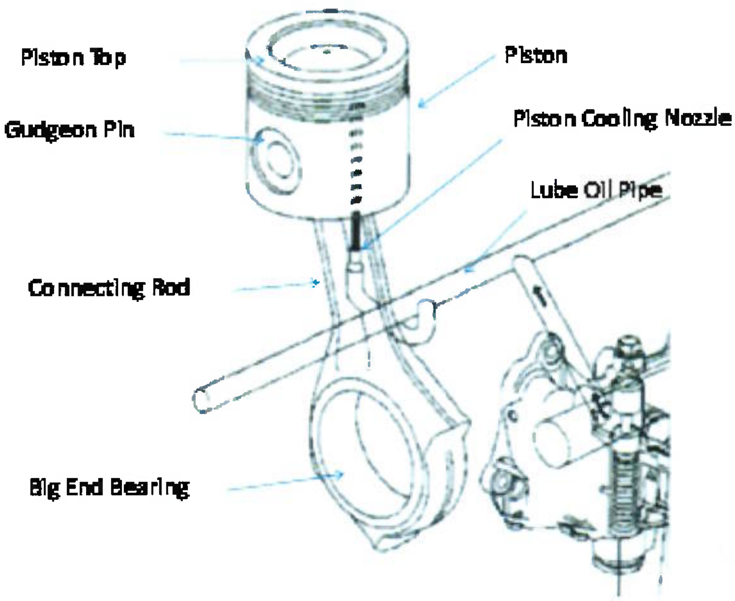
From above, it may be observed that 4-Yearly major maintenance (which includes Half Yearly, Yearly, 2-Yearly Maintenances also) was due in May 2014. However, no record of this major schedule maintenance of engine was available.

**<u>Probable Reasons of Breakdown</u>:**

The major damage has occurred to the Piston and the Liner No. 6. The Piston is made up of Aluminum having a melting temperature of 660 degree C. In this case, the top portion of Piston has melted and the molten material was found stuck up over the Exhaust Valves and also on the broken pieces of Piston Ring.

From the nature of damage caused to the top portion of Piston No. 6, it appears that it may be due to excessive heat generation.

In this engine, the Piston is cooled by Cavity Cooling in which oil is sprayed vertically up in channel in the Piston via Piston Cooling Nozzle in the Cylinder Block. The oil then continues up to a circular channel in the top of Piston and is drained back to the oil pan.




It appears that excessive heat generation on the top of piston may be due to starvation of lube oil which may due to:

- Inadequate supply of lube oil to the top of piston due to chocking of Piston Cooling Nozzle / Piston Channel

- Washing away of lube oil film between Liner and the Piston Rings by Diesel due to improper atomization by Injector Nozzle as no record of Injector Calibration was available.

After the piston seizure, the piston might have broken into pieces and it's Gudgeon Pin became loose in the Connecting Rod small end which in-turn damaged the side wall of the engine block during its free movement inside the engine block.

There might have been some delay in rising of jacket water temperature up to the set point for initiating shut down command causing damage to the engine. (As per report of Yearly Maintenance done in March 2014, simulation for tripping on Jacket water temperature High was carried out at 93 Deg C. However, no record of tripping check on Lube Oil pressure Low was found).

The lube oil sample has been sent to R&D Centre for testing to ascertain whether the engine failure was attributable to any deficiency in its quality.

**Action for Restoration of the DG Set**

The service engineer from the authorized dealer M/s. Rai & Sons visited site on 20.02.2015. As per the report received from the authorized dealer vide e mail dated 21.02.2015 and 23.02.2015(**Flag D**), following major components of the engines are required to be replaced on visual examination:

- Engine Block, as the damaged one is not repairable
- All Cylinder Liners (6 No.)
- All Pistons (6 No.)
- All Main Bearings and Big End Bearings (6 No.)
- Turbocharger (1 No.)
- Fuel Injectors (Numbers to be decided depending upon the condition)

The vendor has estimated the cost of engine rebuilding on preliminary basis as Rs.20 lakh considering replacement of damaged block with one used block which may be arranged by them. However, for preparing firm estimate, the engine is required to be sent to their workshop at Gurgaon where individual parts will be inspected thoroughly after dismantling of engine. The vendor has also indicated the cost of new engine as Rs.22.5 lakh (Approx.).

From above, the repairing cost of the engine works out to about 80-90% of the new engine. However, the residual life and the reliability of repaired engine will be less for a service like emergency power supply in an operating pump station. Hence, it is proposed that we may consider procurement of new engine which may be coupled with the existing Alternator after condemnation of the damaged 15 year old engine. The Alternator has been overhauled recently.

It may be worthwhile to mention here that SMPL Rewari pump station is being shifted to MPPL Rewari station premises with new motor driven pumping units where one 400 KVA DG set is already available and two new 2000 KVA HT DG sets are being installed under SMPL DBN project. The total emergency load requirement has been reviewed by NRPL Rewari and as informed by SIC Rewari, the existing 400 KVA DG set will be sufficient to meet the requirement including new load of SMPL pump station. However, the requirement of new DG set may be reviewed by NRPL Panipat.

**Suggestions for Improvement of Maintenance and Operation Practices:**

- To carry out all maintenances as per schedule
- Feedback of DG running status to be made available in the SCADA.
- Keeping the protective devices in healthy condition by simulation for tripping of DG set for both Lube oil pressure low and Jacket water temperature high tripping from AMF panel and the same to be recorded in maintenance report format.
- During continuous running of DG, parameters (Lube Oil pressure, Jacket water temperature, RPM, load current etc.) to be recorded on hourly basis.
- Technical Literature of DG set and AMF panel to be arranged and made available in the station.

Submitted Please.

05/03/2015
(Rajesh Kumar Mishra)
DPJM (T&I), PLHO

(Tribhuwan Kumar)
SOM, NRPL, Panipat

05.03.2015.
(R.P.Pant)
DGM (MNT), PLHO

**~~DGM (I/c, Maint.)~~**

As brought by committee, the major suspected reason of the failure of the DG set is inadequate supply of lube oil to the top of piston due to chocking of piston cooling nozzle which led to excessive heat generation & washing away the lube oil film between liner & piston ring.

All the above indicates inadequate maintenance. To avoid such type of failure in future corrective actions as suggested by committee are to be strictly followed at each locations.

The report forwarded for appraisal,

9.3.2015
T.K.Bose.

~~GM I/c (O)~~

ED ONSM
17/3/2015

~~DGM (I/c) Maint~~

~~D(M)~~ - may kindly see, before final circulation. Suitable actions needs to be taken for

17/3/2015

- Amending the maintenance manuals for taking routine checks as suggested by the committee
- Ensuring that schedule maintenance has been carried out as per schedule
- Identification of lapses in maintenance due which led to this accident and initiating suitable actions against the persons, who have committed these actions.
- ATR may be submitted on the above in next PMC.

27/5/2015

ED(O)

इंडियनऑयल

**Pipeline Head Office**
**Maintenance Department**

**PL/HO/MNT/01**
**Feb 14, 2015**

**Sub: Investigation of failure of Volvo Penta DG set at SMPL Rewari**

On 13.02.2015, at SMPL, Rewari engine operation was continued at night to meet the increased throughput demand. At 22:00 hrs, the grid power supply failed and to continue engine operation, the 500 KW Volvo Penta DG set was started immediately. At 04:05 hrs on 14.02.2015, the DG set is reported tripped with abnormal sound. After preliminary inspection it has been reported that the liner and engine body block got burst. The engine is not taking manual cranking and the crank shaft is reported to be seized.

Because of the non-availability of power supply, the engine operation at SMPL Rewari got affected due to failure of the DG set.

Considering the severity of breakdown of the DG set, the incident needs to be thoroughly examined to identify the root cause of failure, identify the shortcomings in the maintenance & operation practices if any for improvement and recommendation of action plan for early restoration of the DG set.

In view of the above, a committee of following members is proposed who will visit site for carrying out the investigation of the incident:-

**Mr. R.P. Pant, DGM (Maintenance), PLHO**
**Mr. Trivuwan Kumar , SOM, NRPL, Panipat &**
**Mr. Rajesh Kumar Mishra DPJM(T&I), PLHO**

Scope of Work :-

a) Identification of root cause of failure
b) Suggestion for improvement of Maintenance & operation practices
c) Action plan for early restoration of the DG set

Committee should submit the report by 20.02.2015.

Submitted for approval please.

14/2/15

T. K. Bera
DGM I/C, Maintenance, PLHO

**GM I/C , Operations**

14/2/2015

DGM (I/c) Maint

'A'

## STATION DG OPERATION & TRIPPING CHECKS

Station Name: SMPL Rewari

Date: 06.03.2014

Frequency of maintenance: Yearly

Test Equipment used for Calibration / Testing

| S.No | Name of Instrument | Tag No / Serial Number | Instrument Calibration Validity up to | Accuracy |
|---|---|---|---|---|
| 1 | DMM | NRPL/REW/T&I/06 | 19.02.2015 | |
| 2 | Temp; Calibration Bath | NRPL/REW/T&I/15 | Apr.-2013-14 | |
| | | | | |

Note / Instructions

1. Observation from previous major maintenance report to be considered and fault reported since last major maintenance of the equipment to be taken up for rectification.
2. The acceptable accuracy of field instruments is defined in QAL/T&I/06. In case the field instruments cannot be readjusted to the desired level same is to be recorded and action for replacement of same is to be initiated.
3. The lists of instruments falling under the system are recorded in approved list of instruments QAL/T&I/08.
4. The instruments are to be checked/ calibrated as per the standard test procedures mentioned in the QAL/T&I/16
5. Strike off whichever is not applicable
6. Additional sheet may be used to provide further information, if any.

### A) SEQUENCE FUNCTIONALITY TESTS: NA

| S.No | Description | Status | Feedback status at SCC |
|---|---|---|---|
| 1 | Remote SCC start / stop of DG | NA | NA |

### B) INSTRUMENT TRIPPING TESTS:

| S.No | Inst. Tag | Normal / healthy state | Simulation test at field level i.e. state changed to | Whether state change observed at panel level? |
|---|---|---|---|---|
| 1 | JW-TSH-1 | NC | NO (93°c) | Yes |

REMARKS (IF ANY):

Signatures

Executed by; AJAY  Reviewed by; ______ Parosh  SIC / HOD (T&I); ______ 16/4/14

| Indian Oil | INTEGRATED QUALITY & ENVIRONMENTAL MANAGEMENT SYSTEM MANUAL OF NRPL PANIPAT BASE | QAF: MMM:14<br>REVISION: 0<br>EFFECTIVE DATE: 01.08.2012 |
|---|---|---|

**250 HRS/MONTHLY MAINTENANCE OF DIESEL ENGINES OF EMERGENCY GENERATOR AND F/F ENGINES**

| | |
|---|---|
| **STATION: SMPL, Rewari** | **EQUIPMENT: Emergency Generator** |
| **LOCATION: Generator Room** | **TYPE/SL. NO. TAD16316/2160028657** |
| **PLANNED MONTH: Sep' 14** / **EXECUTION MONTH: Sep' 14** | **RUNNING HRS. – 1497.4** |

**REASONS FOR DELAY (IF ANY): No delay**

| S. NO: | DESCRIPTION | YES | NO |
|---|---|---|---|
| 1. | DRAINED SLUDGE AND CLEANED ELEMENTS OF L.O. EDGE FILTERS. | √ | |
| 2. | INSPECTED AIR FILTER AND CLEANED IF NECESSARY | √ | |
| 3. | IN CASE OF OIL BATH AIR FILTER, CHECKED OIL LEVEL AND CLEANED, IF NECESSARY. | N.A | |
| 4. | CHECKED LUB. OIL FOR CONTAMINATION AND WATER | √ | |
| 5. | DRAINED AIR RECEIVER | √ | |
| 6. | FUEL FILTER DRAINED AND CLEANED ELEMENTS. | √ | |
| 7. | READJUSTED TENSION OF FAN BELTS | √ | |
| 8. | CHECKED FREE MOVEMENT OF THROTTLE LINKAGE | √ | |

**REMARKS: Lube oil replaced and filters changed.**

**WORK CARRIED OUT** FROM 18.09.14 TO 18.09.14

**WORK CARRIED OUT BY**

| NAME: | DESIGNATION | SIGNATURE: for Sumit |
|---|---|---|
| 1. Sumit Sharma | E.A. (M) | NAME: Sumit Sharma<br>DESIGN. E.A. (M) |
| | | COUNTER SIGNED:<br>NAME: Ravindra Kumar<br>DESIGN: O&ME |

**DISTRIBUTION:**
STATION COPY

**PANT, R. P.(आर.पी. पंत)**

**From:** Gautam Banerjee [gbanerjee12@gmail.com]
**Sent:** 23 February 2015 10:46
**To:** PANT, R. P.(आर.पी. पंत)
**Cc:** JHA, J. K. (जे.के.झा); GARG, N.K. (एन.के.गर्ग); pchakravarty@rai-group.com; pkroy@rai-group.com; vschauhan@rai-group.com; prem.d.gserviceengg@gmail.com; raipreximgurgaon@gmail.com
**Subject:** Breakdown report - Volvo Engine- TAD 1631GE Sr N - 2160028657

Dear Mr R P Pant

This further refers to our email of 21st Feb 2015 regarding the visit of our engineer Mr Prem Jha to your site on 20th Feb 2015 for the inspection of the damage Volvo Engine model TAD 1631GE with a serial number - 2160028657 which is installed at your Rewari unit . During the inspection Engine block at cylinder no 6 location found damage & along with its piston which was found in badly damage condition . Also as per the visual inspection of Mr Prem Jha - it is recommended to replace all the 6 nos cylinder liners , 6 nos pistons , all CR & MJ bearings , Turbo charger , some fuel injector etc with new one for reliable performance of the engine .

The engine crank shaft rotation was found free & no other visual damage in the other engine cylinders were found which gives a clear indication that there was no failure in the Lube oil system of the engine .

The probable reason of the failure seems to be due to the failure of the injector of the cylinder no 6 or seizure of the Piston may be due to overloading of the Engine or some leakage problem in valve system of the Cylinder head no 6 , due to which the exhaust temp of the particular damage cylinder head may have gone high . After the piston seizure , the piston breaks into pieces & it's G.pin become loose in the Con Rod small end which in-turns damage the side wall of the engine block during its free rotation in the system .

The several loose pieces of the Pistons & engine block casting got contaminated in the Engine lube oil system & produce scratch marks in the cylinder liner's wall of the other cylinders . Also some particle of the damage piston of cylinder head no 6 may have entered in the turbo charges through the exhaust system due to which turbo chargers rotation is not free , hence the turbo charger also needs to be replaced with new one .

We also request you to give us maintenance schedule of the Engine which was last carried out at your end , so that we can correlate all the information together to conclude the reasoning .

As per our engineer's initial inspection report we have calculated the tentative budgetary cost of the spare parts which he has recommended now & is coming around in the range of
Rs 20 lakhs which apparently makes the repairing option non- viable with the present spare parts replacement recommendation , so have two options now to proceed further in this subject and are as follows :

1. To replace the engine with a new Engine using its old alternator , fuel system , Electrical control system , canopy - **The approx. Cost of this replacement will be around 22.5 Lakhs .**

2. To send the complete engine at our Gurgaon works , so that detailed inspection of the Engine at their component level could be done based on measurement limits recommended by the original engine manufacture & accordingly the final revised spare parts replacement list could be prepared at our end . In case there are some major revision in the spare parts requirement - resulting a substantial cost saving , then again the repairing option can be considered . We confirm that we will not charge anything for this detailed inspection of the components which will carried out at our

Gurgaon works , however the to & Fro transportation , insurance & packing of the engine block from your works to our works will be in your scope . Also note that this option will be only valid in case the engine block which we have located with some customer at present is available with them till we make our final decision on this subject .

We now request you to advise further & accordingly we shall proceed further in this matter .

Thanks & Best Regards

Gautam Banerjee

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**From:** Gautam Banerjee [mailto:gbanerjee12@gmail.com]
**Sent:** Saturday, February 21, 2015 1:28 PM
**To:** 'PANT, R. P.(आर.पी. पंत)'
**Cc:** 'JHA, J. K. (जे.के.झा)'; 'Pchakravarty <pchakravarty@rai-group.com> (pchakravarty@rai-group.com)'; ''pkroy@rai-group.com' (pkroy@rai-group.com)'; 'raipreximgurgaon@gmail.com'
**Subject:** RE: MAINTENANCE OF 1 X 500 KVA VOLVO PENTA DG SET.

Dear Mr Pant

This refers to the telephonic discussion we had today morning regarding the submission of the repairing Estimate of the damage Volvo engine- TAD 1631 which is installed at your Rewari unit .

Based on your confirmation we had deputed our engineer to your site on 20th Feb 2015 for the visual inspection of the damage engine . During the inspection the engine block is found damaged & appears to be beyond repair . As the new engine block is very expensive , **we are trying to locate some good condition Engine block of same model** so that the repairing proposal of the Engine become viable because as per our Engineer's visual inspection report all the 6 nos cylinder liners , 6 nos pistons , all CR & MJ bearings , Turbo charger , some fuel injector etc needs replacement with new & we have our doubt on the viability of the repairing option if we consider all these items .

We hence once again request you to check the possibility to send the complete engine at our Gurgaon works , so that detailed inspection of the Engine components could be done based on measurement limits & accordingly the final replacement requirements / offer could be prepared at our end .

Please advise further .

As required sending you the information related to TAD 1631 GE engine .

Thanks & Best Regards

Gautam Banerjee

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**From:** pk roy [mailto:pkroy@rai-group.com]
**Sent:** Thursday, February 19, 2015 12:36 PM
**To:** 'JHA, J. K. (जे.के.झा)'
**Cc:** 'Lal, Madan (लाल, मदन)'; 'SAMTANI, RAMESH KUMAR ( रमेश कुमार समतानी)'; 'BERA,T.K. (टी.के. बेरा)'; 'PANT, R. P.(आर.पी. पंत)'; pchakravarty@rai-group.com; 'virender Singh Chauhan'; 'Gautam Banerjee'
**Subject:** RE: MAINTENANCE OF 1 X 500 KVA VOLVO PENTA DG SET.

Dear Mr. Jha,

Thanks for your prompt reply and have noted the contents.

Regards-PKRoyChowdhury(+91-9910491036)
**Rai & Sons Pvt Ltd New Delhi.**

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**From:** JHA, J. K. (जे.के.झा) [mailto:JHAJK@INDIANOIL.IN]
**Sent:** Thursday, February 19, 2015 12:18 PM
**To:** pkroy@rai-group.com
**Cc:** Lal, Madan (लाल, मदन); SAMTANI, RAMESH KUMAR ( रमेश कुमार समतानी); BERA,T.K. (टी.के. बेरा); PANT, R. P.(आर.पी. पंत)
**Subject:** RE: MAINTENANCE OF 1 X 500 KVA VOLVO PENTA DG SET.

Dear sir,
As discussed with you, one engineer can be sent from your office at Rewari for one day to assess the extent of damage. Your offer is acceptable to us.

*Thanks and Regards,*

*( K.Jha)*
*DGM, NRPL, Rewari.*
*Cell- 9416201612*
*Phone(O) - 01274-269663*

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**From:** PANT, R. P.(आर.पी. पंत)
**Sent:** 19 February 2015 11:45
**To:** JHA, J. K. (जे.के.झा)
**Cc:** Lal, Madan (लाल, मदन); SAMTANI, RAMESH KUMAR ( रमेश कुमार समतानी); BERA,T.K. (टी.के. बेरा)
**Subject:** FW: MAINTENANCE OF 1 X 500 KVA VOLVO PENTA DG SET.
**Importance:** High

As discussed, the offer of M/s **Rai & Sons Pvt Ltd New Delhi who are the authorized dealer of M/s Volvo Penta is enclosed herewith.**

For information and further needful, please.

Regards,

R.P.Pant
DGM(MNT), PLHO, Noida
Ph.: 0120-2448907

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**From:** pk roy [mailto:pkroy@rai-group.com]
**Sent:** Wednesday, February 18, 2015 7:51 PM
**To:** 'rppant@indianoil.in'
**Cc:** 'Pchakravarty'; virender Singh Chauhan (vschauhan@rai-group.com)
**Subject:** MAINTENANCE OF 1 X 500 KVA VOLVO PENTA DG SET.
**Importance:** High

Ref: Rai/IOCL-Rewari-2015/18.02.2015.

Dear Mr. R.P.Pant,

This has reference to your discussion had with the undersigned on date regarding your installed 1 X 500 KVA Volvo Penta DG Set( Model No TAD1631GE, Serial No 2160028657) is under Break down.

As per the discussion we wish to inform you that we have been appointed by Volvo Penta India Pvt Ltd , as their Authorised Dealer copy of certificate enclosed for your ready reference . As discussed we can undertake the job i.e.. supply of Spare parts and Maintenance of your existing installed Diesel Generating Sets.

We are pleased to submit our offer of engineer service charges as desired by you.

1. We will charge @ Rs 5000.00 per day per engineer.
2. You will provide Skilled/Unskilled labour during the job.
3. You will provide all technical support during the job.
4. You will provide Boarding /Lodging during the stay of our engineer at your site.

**Terms & Conditions:**

**Tax:** We will charge @12.36% Service Tax on total amount.
**Payment:** 100% against completion of job

Hope you will find the same in order and look forward to receiving your favourable reply.

Should you require any further clarification please do not hesitate to contact us if require we will visit to your office for discussion.

Regards-PKRoyChowdhury(+91-9910491036)
**Rai & Sons Pvt Ltd New Delhi.**

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